

Prithvi Chakravarthy

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PROFESSIONAL SUMMARY

Master of Science in Computer Science student with over three years of professional full-stack engineering experience. Adept at building scalable web applications, designing distributed systems, and integrating applied artificial intelligence into functional software. Proven ability to translate complex technical requirements into accessible digital solutions, with a strong commitment to clean, modular architecture.

EDUCATION

DePaul University <i>Master of Science in Computer Science</i>	Chicago, IL Sept. 2025 – Present
BITS Pilani, WILP <i>Postgraduate Certificate in AI/ML</i>	India Oct. 2023 – Oct. 2024
VIT Bhopal <i>Bachelor of Technology in Computer Science</i>	India Jun. 2018 – Jun. 2022

EXPERIENCE

Designare Solutions Pvt Ltd <i>Full Stack Developer</i>	Mumbai, India March 2022 – August 2025
- Developed and deployed full-stack applications—including the Deepstory, Leisurely, and Iqgeo platforms—using React, Node.js, Python, and PostgreSQL, ensuring scalable and maintainable codebases. - Designed distributed APIs and optimised data pipelines, reducing query time by 25%. - Managed production deployments on Google Cloud, maintaining 99.9% uptime while implementing secure user authentication and role-based access control (RBAC). - Collaborated with stakeholders to define requirements and deliver on time.	

PROJECTS

AI Journal OS <i>React, TypeScript, Rust, Tauri, gRPC</i>	
- Engineered a privacy-first, local-offline AI journaling prototype utilizing Tauri, Rust, and React. - Architected a two-stage AI privacy pipeline that intercepts user prompts via a background Rust thread, utilizing the candle framework to locally redact PII before routing queries to the Groq cloud API. - Built a secure, encrypted-at-rest local vault using AES-256 (magic.crypt) and developed a custom hybrid sync engine to push ciphertext securely to a Supabase PostgreSQL cloud database.	
Space Invaders Game Engine <i>C#, Azul Engine, IrrKlang</i>	
- Engineered a high-performance Space Invaders clone using C# and the Azul engine, strictly adhering to SOLID principles and implementing 14+ GoF design patterns to ensure a modular and decoupled architecture. - Optimized memory management by architecting a universal Object Pool system (Template pattern), eliminating runtime heap allocation spikes and leveraging Proxy/Flyweight patterns to reduce GPU memory overhead for sprite rendering. - Developed a decoupled collision detection system using Visitor and Observer patterns for double-dispatch resolution, and managed interval-based game events via a Command pattern paired with a Priority Queue.	
Stable Diffusion Image Generation Model <i>Python, PyTorch, Jupyter</i>	
- Developed a latent diffusion model using PyTorch to generate high-fidelity images from text prompts, focusing on efficient tensor operations and model architecture. - Implemented and optimized U-Net and attention mechanisms to improve inference speed and generation quality. - Managed end-to-end data preprocessing and training pipelines, ensuring numerical stability and convergence across large datasets.	

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Rust, Java, Kotlin, C#
Frameworks/Libraries: React, Node.js, Tauri, PyTorch, TensorFlow, Pyramid, TailwindCSS
Databases: PostgreSQL, MySQL, Oracle
Tools: Google Cloud, GitHub, Docker, Jenkins, CI/CD, REST APIs, Jest